

Theorem 1. Let $h: \Sigma^* \rightarrow \Gamma^*$ be a homomorphism.

if $A \subseteq \Sigma^*$ is regular then $h(A)$ is regular.

Proof. α is a regular expression s.t. $L(\alpha) = A$.
 $\downarrow$ over Σ .

To show: Construct α' s.t. $L(\alpha') = h(A)$.

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α' - Replace each letter (or symbol) $a \in \Sigma$ in α with the string $h(a) \in \Gamma^*$.

Formal defn. of α' - By induction on the structure of α .

$$a' = h(a), a \in \Sigma.$$

$$\phi' = \phi$$

$$(\beta + \gamma)' = \beta' + \gamma'$$

$$(\beta \gamma)' = \beta' \cdot \gamma'$$

$$(\beta^*)' = \beta'^*$$

$\downarrow$ over Σ .

Claim. For any regular expression β : $L(\beta') = h(L(\beta))$

Corollary: $L(\alpha') = h(A)$

Lemma 2. For $C, D \subseteq \Sigma^*$, $h(CD) = h(C)h(D)$.

Lemma 3. For a family of subsets $C_i \subseteq \Sigma^*$, $i \in I$,

$$\text{we have } h\left[\bigcup_{i \in I} C_i\right] = \bigcup_{i \in I} h(C_i)$$

Exercise: Proof of Lemma 2 and Lemma 3.

To prove: For any β , $L(\beta') = h(L(\beta))$

Base Case:

$$L(a') = L(h(a)) = \{h(a)\} = h(\{a\}) = h(L(a))$$

$$L(\phi') = L(\phi) = \phi = h(\phi) = h(L(\phi))$$

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Induction Step: Operators $+$, $\cdot$, $*$

$$L((\beta + \gamma)') = L(\beta' + \gamma') \quad [\text{Defn of } ']$$

$$= L(\beta') \cup L(\gamma') \quad [\text{Defn. of } +]$$

$$= h(L(\beta)) \cup h(L(\gamma)) \quad [\text{Induction hypothesis}]$$

$$= h(L(\beta) \cup L(\gamma)) \quad [\text{Lemma 3}]$$

$$= h(L(\beta + \gamma)) \quad [\text{Definition of } +].$$

For concatenation, the proof is similar to $+$. Use Lemma 2.

$$L(\beta^{*'})$$

$$= L(\beta'^{*}) \quad [\text{Definition of } ']$$

$$= L(\beta')^* \quad [\text{Defn. of regular expression operator } *].$$

$$= h(L(\beta))^* \quad [\text{Induction hypothesis}]$$

$$= \bigcup_{n \geq 0} h(L(\beta))^n \quad [\text{Defn. of Set operator } *]$$

$$= \bigcup_{n \geq 0} h(L(\beta)^n) \quad [\text{Lemma 2}].$$

$$= h\left(\bigcup_{n \geq 0} L(\beta)^n\right) \quad [\text{Lemma 3}]$$

$$= h(L(\beta)^*) \quad [\text{Defn of Set operator } *]$$

$$= h(L(\beta^*)) \quad [\text{Defn of regular expression operator } *].$$

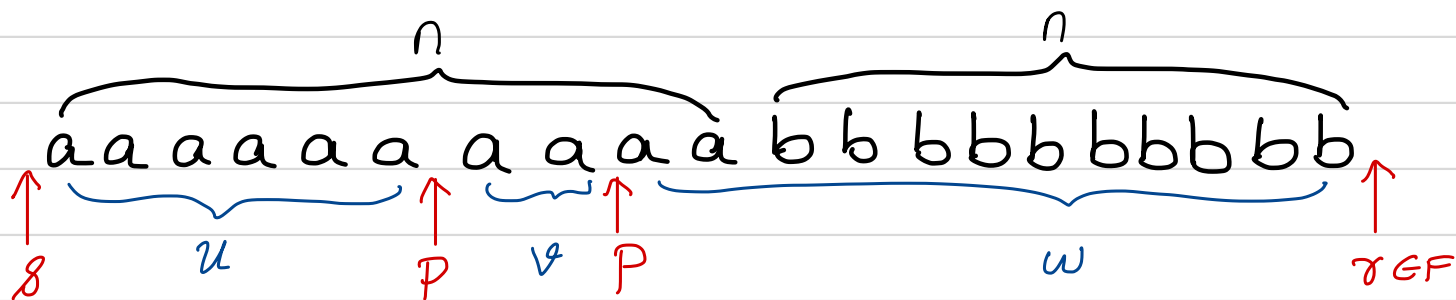
$$B = \{a^n b^n \mid n \geq 0\} = \{\epsilon, ab, aabb, aaabbb, \dots\}$$

Question. Is B regular?

Suppose B is regular. Then there is a DFA M s.t. $L(M) = B$

Let k be the number of states of M .

Consider an arbitrary string $a^n b^n$ where $n > k$.



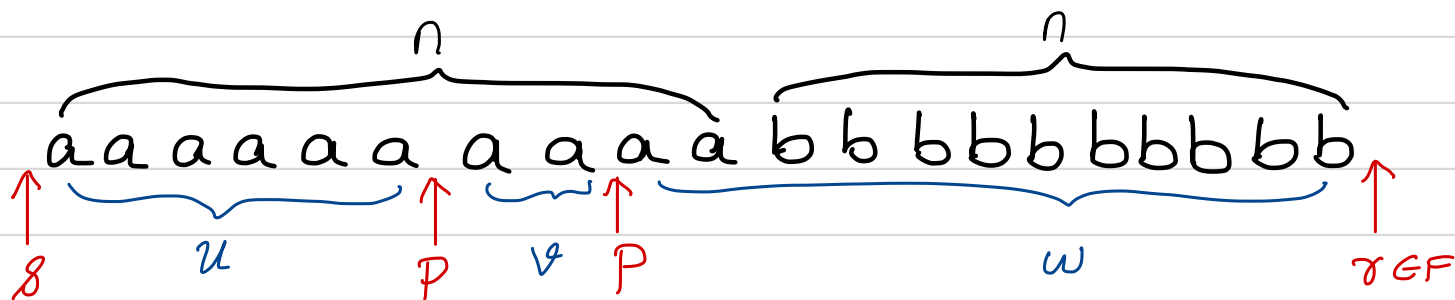
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There exists a state p in M such that

$$\hat{\delta}(\delta, u) = p; \quad \hat{\delta}(p, v) = p \text{ where } |v| = j > 0$$

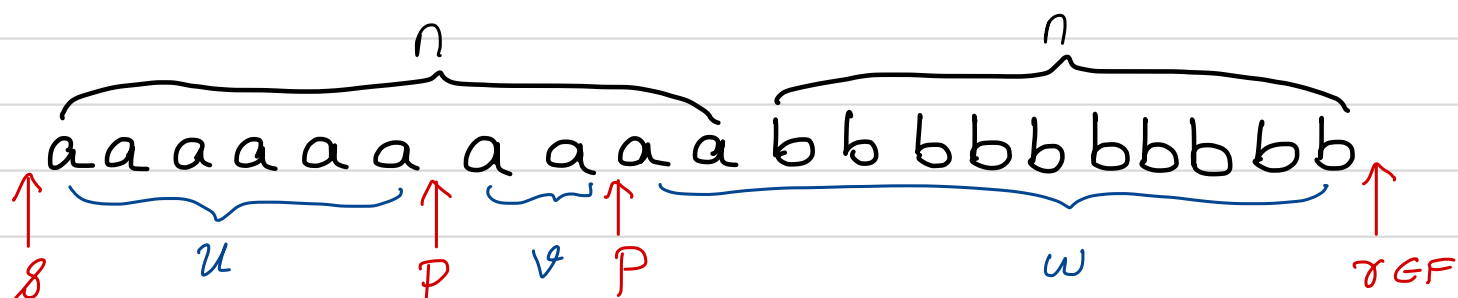
$$\text{and } \hat{\delta}(p, w) = \delta \notin F$$

Now consider the run of M on input uw .

$$\hat{\delta}(\delta, uw) = \hat{\delta}(\hat{\delta}(\delta, u), w) = \hat{\delta}(p, w) = \delta \notin F$$

$$uw \in L(M) \text{ but } uw = a^{n-|v|} b^n \notin B$$

Consider an arbitrary string $a^n b^n$ where $n > k$.



Consider the string $uv^2w = a^{n+|v|} b^n$ and the run of M on uv^2w .

$$\begin{aligned}
 \hat{s}(s, uv^2w) &= \hat{s}(\hat{s}(\hat{s}(\hat{s}(s, u), v), v), w) \\
 &= \hat{s}(\hat{s}(\hat{s}(p, v), v), w) \\
 &= \hat{s}(\hat{s}(p, v), w) \\
 &= \hat{s}(p, w) = \gamma \in F.
 \end{aligned}$$

Thus $uv^2w \in L(M)$

But $uv^2w = a^{n+|v|} b^n \notin B$.

Contradicts our assumption that $L(M) = B$

Let A be a regular Set. Then the following property (P) holds for A .

(P) { There exists $k \geq 0$ such that for any string x, y, z with $xyz \in A$ and $|y| \geq k$ there exists strings u, v, w s.t. $y = uvw$, $v \neq \epsilon$ and for all $i \geq 0$, the string $xuv^i w z \in A$.

Pumping Lemma.

Let A be a regular Set. Then the following property (P) holds for A .

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Contrapositive of (P): Suppose A satisfies the following:

($\neg P$) { For all $k \geq 0$ there exists strings x, y, z such that $xyz \in A$, $|y| \geq k$ and for all u, v, w with $y = uvw$ and $v \neq \epsilon$, there exists $i \geq 0$ s.t. $xuv^i w z \notin A$.

Then A is not regular.

Suppose A satisfies the following:

(7P)

For all $k \geq 0$ there exists strings x, y, z such that $xyz \in A$, $|y| \geq k$ and for all u, v, w with $y = uvw$ and $v \neq \epsilon$, there exists $i \geq 0$ s.t. $xu v^i w z \notin A$.

Then A is not regular.

Example. $A = \{a^n b^m \mid n \geq m\}$.

Claim. A is not regular.

By pumping lemma, suffices to show that A satisfies 7P

Consider any $k \geq 0$, let $x = a^k$, $y = b^k$ & $z = \epsilon$.

Then $xyz \in A$.

Suppose A satisfies the following:

(7P)

For all $k \geq 0$ there exists strings x, y, z such that $xyz \in A$, $|y| \geq k$ and for all u, v, w with $y = uvw$ and $v \neq \epsilon$, there exists $i \geq 0$ s.t. $xu v^i w z \notin A$.

Then A is not regular.

Example. $A = \{a^n b^m \mid n \geq m\}$.

Claim. A is not regular.

By pumping lemma, suffices to show that A satisfies 7P

Consider any $k \geq 0$, let $x = a^k$, $y = b^k$ & $z = \epsilon$.

Then $xyz \in A$. Consider any split of $y = uvw$ with $v \neq \epsilon$. Say $y = \underbrace{b^j}_u \underbrace{b^m}_v \underbrace{b^n}_w$. $\therefore k = j + m + n$

Let $i = 2$. $xu v^2 w z = a^k b^j b^m b^m b^n$

$$= a^k b^{j+2m+n}$$

$$= a^k b^{k+m} \notin A$$